

Filip Mučnjak

Software engineer

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Software engineer with 5+ years of experience crafting high-quality digital solutions. Specializing in .NET and modern frontend frameworks, I build scalable web applications and architect robust backend systems. Currently pursuing a Master's in Computer Engineering while running a software development business and delivering expert technical consulting.

Experience

Factory X · Pribislavec, Croatia

Construction Workforce Management Platform · OCT 2025 - MAY 2026

Engineered a custom workforce management platform for the construction sector, delivering features for on-site coordination, resource tracking, and field operation digitization to optimize core business processes.

Vue.js · .NET Web API · Aspire · MS SQL Server · Quasar · SignalR · Identity · EF Core · Quartz.NET · Azure Storage

King ICT · Buzin, Croatia

User Management System · MAY 2025 - AUG 2025

Developed a comprehensive user management web application featuring role-based access control and lifecycle management to support a large-scale, microservice-based technical protection ecosystem.

.NET Blazor · Syncfusion · Strawberry Shake · bUnit · xUnit · Moq

Visitor Announcement System · FEB 2025 - MAY 2025

Designed and implemented a web application and REST API integrated with a third-party SDK to manage visitor announcements within a comprehensive technical security platform.

.NET Blazor · .NET Web API · MS SQL Server · Dapper · MudBlazor

Infinum · Zagreb, Croatia

Creator – Fan Social Media Platform · FEB 2024 – FEB 2025

Co-developed a creator-fan social media platform featuring exclusive content feeds and direct messaging, while implementing comprehensive admin and agent dashboards for seamless system and creator management.

.NET MVC · .NET Web API · PostgreSQL · jQuery · Bootstrap · Azure AD B2C · Firebase · EF Core · SendBird · Azure Storage · xUnit · NSubstitute

Sport Events Management System · OCT 2023 – FEB 2024

Built a robust REST API providing backend support for a mobile sports event management platform that allows users to seamlessly host events, join activities, and track performance.

.NET Web API · PostgreSQL · Azure AD B2C · EF Core · xUnit · Moq

Blood Donation Coordination Hub · APR 2023 – OCT 2023

Developed an admin dashboard and REST API for a blood donation platform that streamlines user scheduling, history tracking, and donation center discovery.

.NET Blazor · .NET Web API · PostgreSQL · Bootstrap · Identity · EF Core · Firebase · SendGrid · xUnit · Moq

Installation Management Network · MAY 2022 – APR 2023

Engineered an admin dashboard and REST API to manage projects, track work orders, and facilitate connections between manufacturers and retailers for retail installation resources.

.NET MVC · .NET Web API · PostgreSQL · jQuery · Bootstrap · Identity · EF Core · Firebase · SendGrid · Azure Storage · xUnit · Moq

Enterwell · Zagreb, Croatia

Conference Device Management Tool · MAR 2022 – MAY 2022

Developed a desktop application connected to a central web dashboard to ensure streamlined monitoring and coordination of conference equipment during events.

.NET WPF · MS SQL Server · SignalR

Technical Security Solution · FEB 2021 – MAR 2022

Integrated web and desktop dashboards to efficiently manage and monitor a comprehensive security platform featuring IoT devices, real-time video streaming, and access control units.

React.js · .NET MVC · .NET WPF · .NET Web API · MS SQL Server · Material UI · jQuery · Bootstrap · Identity · EF · SignalR · xUnit · Moq

Skills

Frontend

PROGRAMMING LANGUAGES

JavaScript · TypeScript · C# · HTML · CSS

FRAMEWORKS

React.js · .NET Blazor · Vue.js · .NET MVC · .NET Razor Pages

LIBRARIES

Tailwind CSS · Material UI · MudBlazor · Bootstrap

TESTING

xUnit · NSubstitute · Selenium

Architecture

SYSTEM

Modular monolith · Microservices · Event-driven

SOFTWARE

Clean arch. Domain driven design · REST · Layered arch.

Paradigms

DESIGN PATTERNS

DI · CQRS · MediatR · Repository · Unit-of-work · Factory · Singleton

DESIGN PRINCIPLES

SOLID · KISS · YAGNI · DRY

METHODOLOGIES

Agile · Scrum · Kanban

Backend

PROGRAMMING LANGUAGES

C#

FRAMEWORKS

.NET Minimal API · .NET Web API

LIBRARIES

EF Core · Identity · MediatR · SignalR · Azure Storage · Firebase

TESTING

xUnit · NSubstitute · Moq

Infrastructure

DEVOPS

GitHub · GitLab · Aspire · Docker · Azure · SonarQube

DATABASE

SQL · PostgreSQL · MS SQL Server

Tools

DEVELOPMENT

Rider · WebStorm · DataGrip · VS Code · Postman · Claude Code

PRODUCTIVITY

Notion · Google Workspace · Figma

OPERATING SYSTEMS

MacOS · Linux · Windows

Education

Professional Master in Computer Engineering

2022 - PRESENT

Algebra Bernays University · Zagreb, Croatia

Advanced C#, Java, Python & JavaScript · Software Design · DevOps · Data Analysis · Critical Thinking

Bachelor in Software Engineering

2017 - 2022

Algebra University College · Zagreb, Croatia

Object-Oriented Programming · Web Technologies · Mobile Development · Computer Networks · Project Management

Technician for Mechatronics

2012 - 2016

Fran Bošnjaković Mechanical Engineering School · Zagreb, Croatia

Programming in C · Microcontrollers · AutoCAD Design

Certificates

Software Development Fundamentals · Microsoft Technology Associate

Database Administration Fundamentals · Microsoft Technology Associate

Introduction to Programming using HTML and CSS · Microsoft Technology Associate